

Swetha Vaidyanathan Sriram

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EDUCATION

The Pennsylvania State University, Pennsylvania

Master of Science, Computer Science

2025 – 2027 | CGPA: 4/4

Madras Institute of Technology, Anna University, India

Bachelor of Engineering, Computer Science and Engineering

2020 – 2024 | CGPA: 3.94/4

Indian Institute of Technology, Madras, India

Bachelor of Science, Data Science & Programming (Hybrid)

2020 – 2024 | CGPA: 3.73/4

SKILLS

Languages & Database – Python, Java, C++/C, JavaScript, C#, Bash, , OracleSQL, MySQL, PostgreSQL, MongoDB, Redis, SQLite

Web Technologies – Flask, ReactJS, Vue.js, Next.js, TypeScript, Node.js, HTML5, CSS3, REST APIs, JSP, FastAPI

AI & ML – LangSmith, LangChain, LangGraph, Scikit-Learn, TensorFlow, Keras, Tableau, OpenCV, PyTorch

Tools & Platforms – Oracle APEX, Celery, ChromaDB, Jenkins, Docker, Kubernetes, Git, GCP, OCI, AWS

WORK EXPERIENCE

Research Assistant | Pennsylvania State University

01/2026 – 05/2026

- Accelerated molecular dynamics simulations by 15%, analyzing 10,000+ trajectories across 50 protein complexes using **C#** and **MATLAB**.

Associate Applications Developer | Oracle Pvt. Ltd.

08/2024 – 08/2025

- Enhanced an enterprise-scale release management platform (100+ users across APAC and EMEA) by **designing and deploying 2 new workflows** using **PL/SQL, JavaScript, and Oracle APEX**, reducing manual scheduling (~20% improvement).
- Designed and developed a **Jira** ticket dashboard (150+ DAUs), reducing weekly status meetings by ~20% and accelerating ticket resolution time by 5%.
- Contributed to **3 Oracle SDK releases**, implementing core features, bug fixes and ensured on-schedule delivery of enterprise updates.
- Resolved critical bugs in **Java** and developed **JUnit** tests, improving system reliability and reducing release-related delays by ~10%.
- Analyzed and optimized **Oracle Cloud Infrastructure (OCI)** logs to identify and eliminate redundant entries.

Project Intern | Oracle Pvt. Ltd.

01/2024 – 07/2024

- Automated** regression **testing flows** with Oracle Utilities Testing Accelerator (**OUTA**) and **Selenium**, boosting test coverage by ~30% and reducing manual QA effort by ~40%
- Built and deployed an energy forecasting model adopted by 10+ data science team members, improving prediction accuracy by ~20% over previous model using **Oracle ADB, Python, Java and Selenium**

ACADEMIC PROJECTS

CodeBase Intel - Code Intelligence Platform

- Built a polyglot tree-sitter AST parser that models cross-file dependencies in **PostgreSQL** (Supabase) with recursive CTEs and renders an interactive graph via **React Flow + Vite**; async ingestion offloaded to **Celery + Redis**
- Enabled semantic code search and **RAG**-based Q&A using **ChromaDB + OpenAI**; deployed to **GCP Cloud Run** with multi-stage **Docker** builds, **Terraform IaC**, and **GitHub Actions CI/CD**

Medical Research Multi-Agent System

- Orchestrated a 3-agent **LangGraph** pipeline (Retriever, Fact-Checker, Synthesizer) querying PubMed and arXiv in real time, embedding results into **ChromaDB** for semantic retrieval, and returning evidence-graded answers with citations
- Instrumented full agent tracing via **LangSmith**; surfaced results through a streaming **FastAPI** backend and **Next.js 15 + TypeScript** frontend

ConflictMind - AI Agent that Remembers You

- Engineered a 3-step debate pipeline using **Google ADK + Gemini 2.5 Flash (Vertex AI)** that detects contradictory memories via vector similarity + LLM classification, resolves them through structured debates, and persists unified memory to MongoDB Atlas
- Integrated a **MongoDB MCP** server for dynamic agent memory I/O; containerized with **Docker** and deployed to **GCP Cloud Run** via **Flask** backend